

Department of Mechanical Engineering

Six-Month Industry Internship Seminar-I

Session: 2026-2027

Date: 18/08/2026

Industry Name: Lemken India Agro Equipment Pvt Ltd

Project: Industrial Energy Monitoring & Analytics System (IIOT)

Presented By:

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Batch: A5

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GHRCE, Nagpur

Industry Guide:

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Designation: Plant

Maintenance Head

Industry Description



About the Industry: LEMKEN operates in the agricultural machinery and farm mechanization industry. It manufactures advanced agricultural implements such as ploughs, cultivators, power harrows, mulchers, and subsoilers.

Historical Background: LEMKEN was founded in 1780 in Germany and has over 245 years of experience in agricultural technology. LEMKEN India was incorporated in 2010 as a subsidiary of LEMKEN GmbH and began manufacturing in Nagpur.

Industry Description

Vision: To become India's most respected premier agricultural implement manufacturer by providing quality products and excellent customer service.

Mission: To improve farmers' productivity and income through reliable, durable and user-friendly mechanized agricultural solutions

Location: D-59, MIDC, Butibori, Nagpur, Maharashtra – 44110

Annual Turnover: 130 Crores

Number of Employees: 200-300

Products:

- Juwel 6 M & Opal Series – Reversible Ploughs
- Achat 70 & Kyanite 7 – Tyne Cultivators
- Perlite 5 – Power Harrow
- Melior – Subsoiler
- Spinel – Mulcher

Mentor Details:

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Problem Identification

The Need for Smarter Industries:

As technology moves toward **greater efficiency, accuracy, and automation**, industries need to evolve beyond conventional mechanical systems. Mechanical power alone is **no longer enough** - the integration of IoT, electronics, software, and AI can bring the **next level of industrial intelligence**.

The Idea Behind IEMAS:

During my internship interview, I was asked **whether energy monitoring across the plant could be automated**. My answer was a confident **“Yes.”** This led to the idea of developing a system that could **automate data collection, reduce manual calculations, and make energy consumption easier to understand through visualization and analytics**.

Industry Challenges:

1. **Manual Calculations** – Energy readings and consumption calculations require manual effort.
2. **Limited Data Visualization** – No centralized dashboard to easily understand energy trends and patterns.
3. **Limited Accessibility** – Energy data is not easily accessible from a centralized location when required.
4. **Data Privacy & Security** – Sensitive industrial data should not be exposed to external AI services.
5. **Private AI Requirement** – An organization-specific/private LLM layer can provide AI insights while maintaining data confidentiality and company policies.

“If we can automate production, why should energy monitoring remain manual?”

Project Description

● IEMAS - Industrial Energy Monitoring & Analytics System (IIOT) ●

OBJECTIVE

- To develop an **IoT-based system** for **real-time monitoring and analysis** of industrial energy consumption.
- To collect meter readings through **RS485 Modbus**, store them securely, and visualize **machine-wise and PMCC-wise consumption**.
- To use **AI insights** to identify **abnormal consumption, energy waste**, and opportunities for reduction.
- To create a **scalable solution** for **plant-wide energy monitoring**.
- To Help Industry to **save on Energy Bills** by optimizing energy planning.
- Access to **Historical data** in visualized formats.

SCOPE

- Monitor **energy consumption** from **industrial energy meters** and collect readings through an **edge device**.
- Provide a **web dashboard** with **live data, historical trends, machine-wise/PMCC-wise analysis**, and **reports**.
- Enable **future expansion** to **multiple meters, plant-wide monitoring, alerts, predictive insights**, and **industrial-system integration**.

BENEFITS

- Reduces **manual meter checking** and dependency on **handwritten records**.
- Helps quickly identify **high energy consumption, idle loads, abnormal spikes**, and **energy wastage**.
- Enables **data-driven decisions, better energy management**, and **potential reduction in operating costs**.

Abstract

Industrial Energy Monitoring & Analytics System (IEMAS) is an IoT-based solution designed to **centralize and simplify** industrial energy monitoring. The system collects energy readings from industrial meters using **RS485 Modbus** communication, stores the data, and presents it through a **web-based dashboard**. **AI-powered insights** help users understand consumption patterns and identify potential energy wastage.

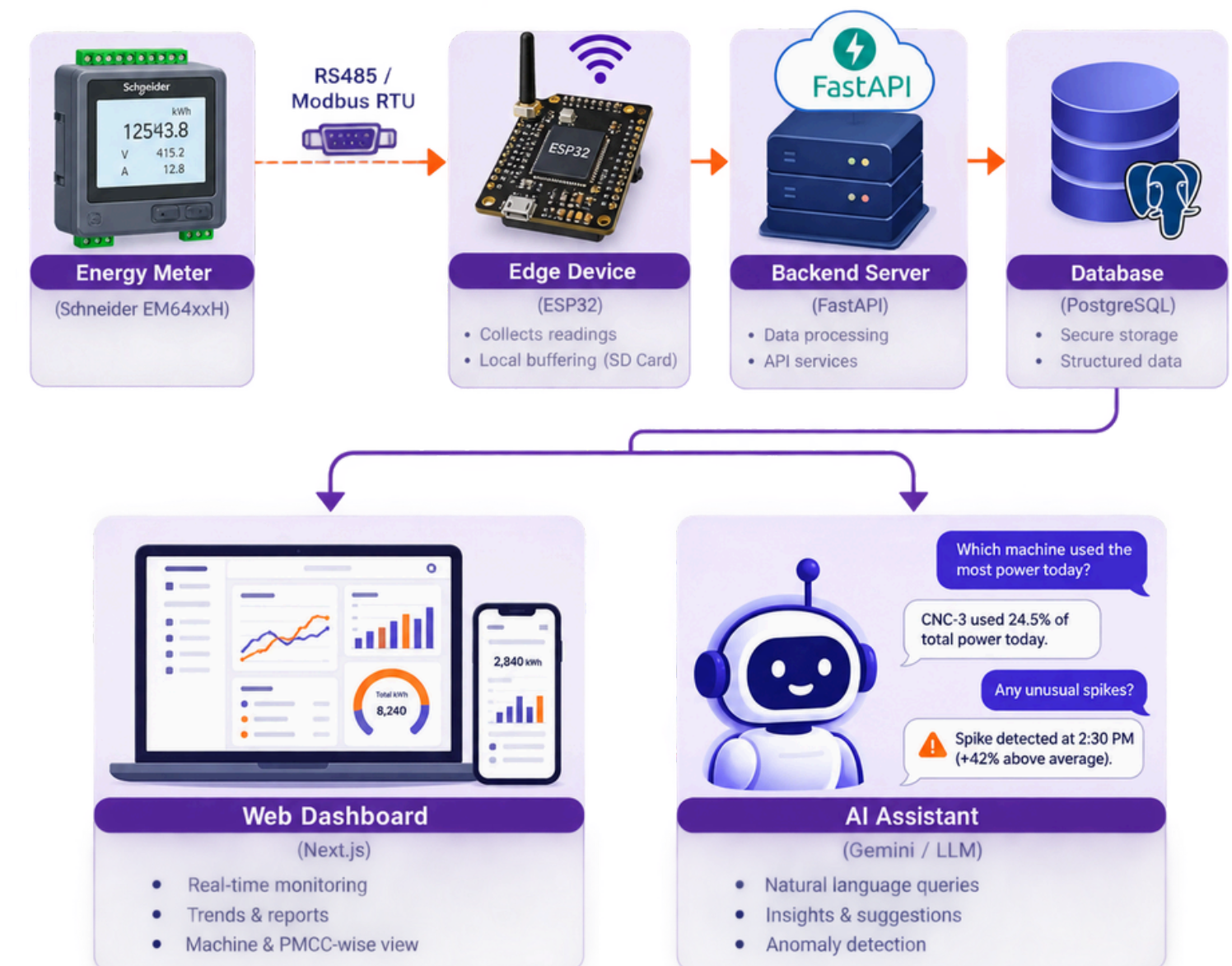
Expected Impact:

- Reduced manual effort
- Faster identification of energy wastage
- Better data-driven energy management

Key Highlights:

1. **Real-time Energy Monitoring** from industrial meters
2. **Machine & PMCC-wise Analysis** through interactive dashboards
3. **Historical Data & Reports** for energy consumption tracking
4. **AI-powered Insights** for trends and abnormal consumption
5. **Local Data Buffering** when internet connectivity is unavailable
6. **Scalable Architecture** for future plant-wide deployment

System Overview



Hardware & Software Specification

Hardware Specifications

Component	Specification / Purpose
Energy Meter	Schneider EM64xxH – industrial energy measurement
Edge Device	ESP32 – collects meter readings
Communication	RS485 + Modbus RTU
Local Storage	SD Card – temporary data buffering
Connectivity	Wi-Fi / network connection for data transmission

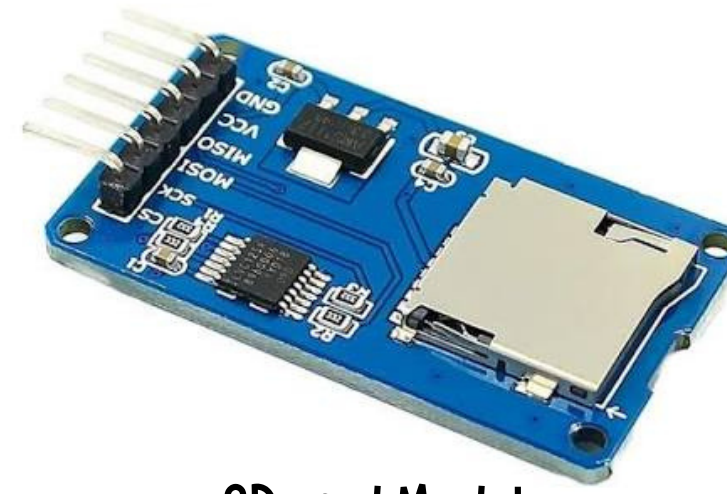
Software Specifications

Component	Technology / Purpose
Frontend	Next.js + Tailwind CSS + shadcn/ui
Visualization	Apache ECharts – energy trends & analytics
Backend	FastAPI – API & data processing
Database	PostgreSQL – structured energy data
Authentication	Supabase Auth
AI Layer	LLM for Insights
Monitoring	Prometheus + Grafana
Deployment & CI/CD	Nginx + Linux VM + GitHub Actions

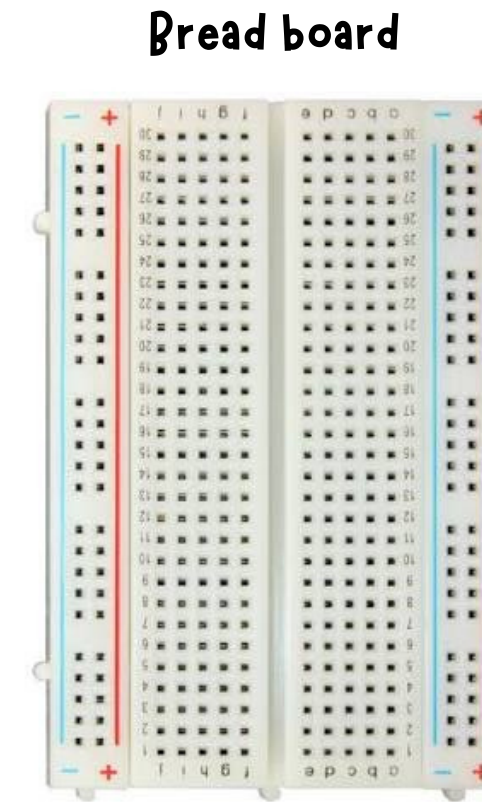
Hardware Components



Schneider Electric
EM64xxH series meter

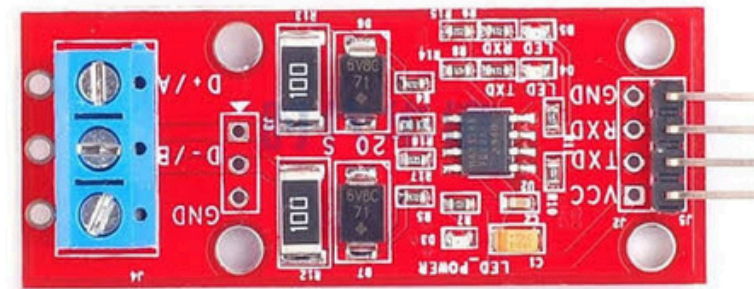


SD card Module



Bread board

ESP32 Devkit v1
Microcontroller

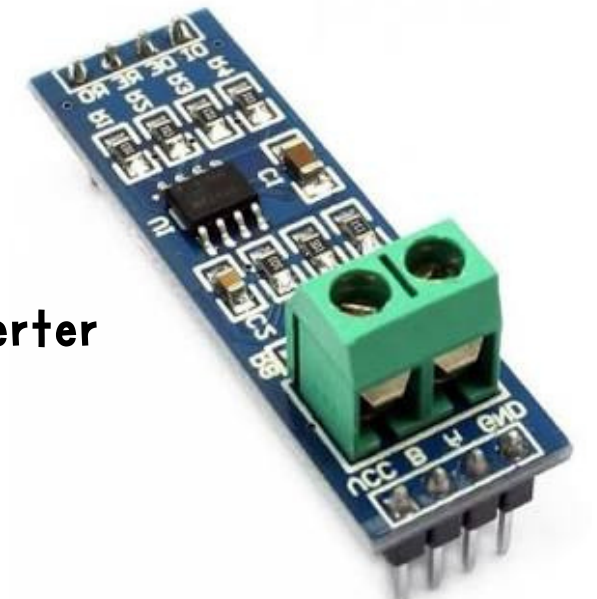


Max485 to ttl converter
Automatic



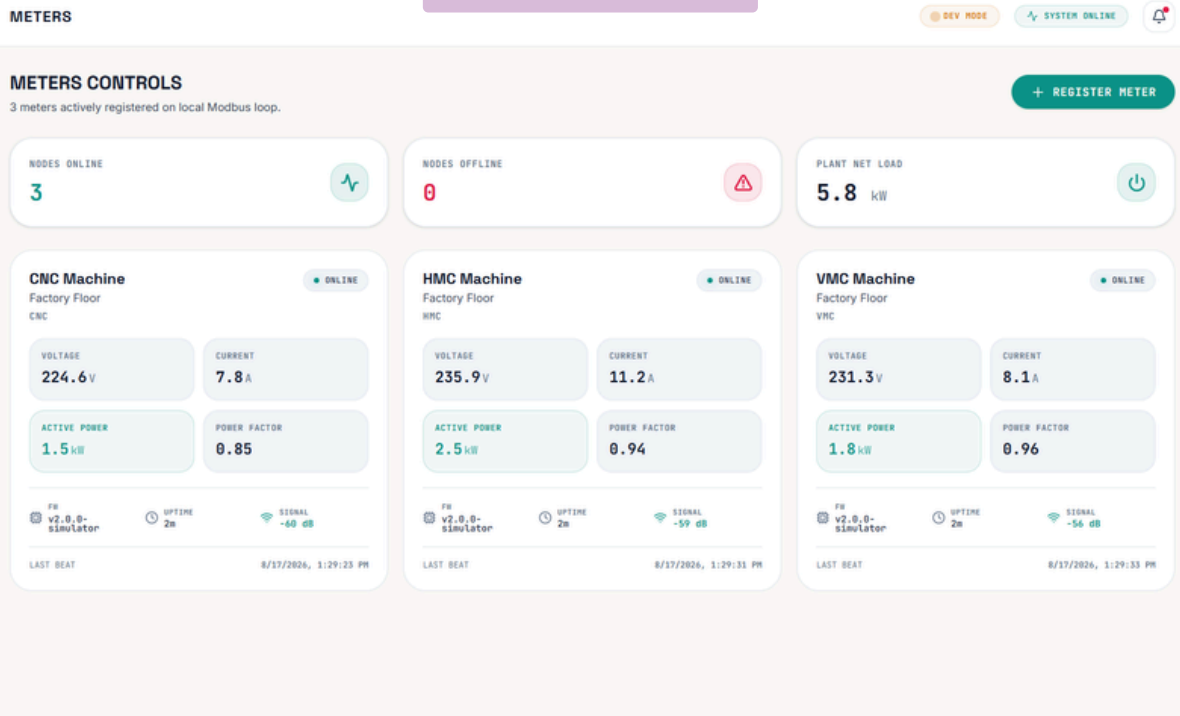
Max485 to Usb Converter

Max485 to ttl converter
Manual



Software Overview

Dashboard



Thank You!